

Chapter 32 — Embryology of the Human Genitourinary Tract

Chapter Notes — source pp. 584–607 (partial coverage)

Partial coverage — source pp. 584–607 only. These notes cover the chapter from its opening through external genitalia, prostate and seminal vesicle, gonads and gonadal descent, Müllerian structures, and the first part of bladder and ureteral development. The supplied text ends mid-sentence on p. 607 (ectopic vas deferens). The remainder of the chapter — the rest of bladder and ureteral development, kidney development, any later tables and Key Points, and the closing section — was not supplied and is not covered here.

- **Two interrelated systems.** The urogenital system has a urinary and a reproductive part, **most closely linked early in development.**
- **Timing.** Development **begins in the first trimester and is mostly complete by the end of the second trimester.**
- **Carnegie collection.** Documents human embryos **from fertilisation to the indifferent stage, at about 8–9 weeks' gestation.**
- **Sexual divergence.** After 8–9 weeks, **androgens drive male and female urogenital structures apart.**
- **Clinical relevance.** Normal development is the **basis for understanding congenital anomalies** and the first step in planning reconstruction.

DEVELOPMENT OF THE EXTERNAL GENITALIA

Genetic and Hormonal Control

- **Indifferent stage.** At 8–9 weeks the external genitalia consist of an **indifferent genital tubercle.**
- **The gonad comes first.** External genital differentiation **depends on prior gonadal differentiation.**
- **SRY.** Gonadal sex is set by the **SRY (sex-determining region Y) region on the short arm of the Y chromosome**, whose downstream cascade directs testicular development.
- **Accessory-structure genes** lie on the **X chromosome and the autosomes**, not only the Y.
- **Leydig cells** appear at **about 8 weeks** and begin secreting testosterone.
- **Targets.** Testosterone and the more potent **dihydrotestosterone (DHT)** masculinise the **genital tubercle, the Wolffian duct and the urogenital sinus (UGS).**
- **Type 2 5 α -reductase** converts testosterone to DHT, is **expressed in the UGS and genital tubercle**, and is required for normal penile and prostatic development.

Masculinisation Sequence

- **Order:** increased anogenital distance (first sign) → **elongation of the genital tubercle** → **formation of the urethral plate** within it.
- **The genital tubercle** is the precursor of **both penis and clitoris** and contains all three germ layers.

GERM-LAYER ORIGIN OF THE GENITAL TUBERCLE

Germ layer	Derivative
Ectoderm	Skin of the phallus and the prepuce
Mesoderm	Corporal bodies (erectile tissue)
Endoderm	Urethral plate → penile urethra, to near the distal tip of the future glans

- **Pattern.** Each layer gives one tissue compartment; the **urethra is endodermal along its whole length** — the basis of the modern view of the glanular urethra.

Penile Urethra — the "Double Zipper"

- **Opening zipper (canalisation).** The urethral plate **canalises proximal → distal** into a **wide, diamond-shaped urethral groove** bounded by the urethral folds.
 - Driven by **cell proliferation (Ki67-positive)**; **no apoptosis** (caspase-3-negative).
- **Closing zipper (fusion).** The **medial epithelial edges of the urethral folds fuse in the ventral midline** → tubular penile urethra and **median penile raphe**.
 - A dense **vascular and neural network** lies at the fusion site.
- **Timetable:**
 - **11–12 weeks:** coronal sulcus separates glans from shaft.
 - **16 weeks:** urethral folds completely fused along the shaft.
 - **18 weeks:** penile and urethral development essentially complete.
 - **20 weeks:** normal fetal ventral curvature (**chordee**) has resolved.

SHAFT URETHRA VS GLANULAR URETHRA — TWO MECHANISMS

	Penile shaft urethra	Glanular urethra
Mechanism	Canalisation, then fusion of the urethral folds	Direct canalisation of the urethral plate , with epithelial remodelling inside the glans
Origin of epithelium	Endoderm (UGS / urethral plate)	Endoderm (UGS / urethral plate) — for most of its length
Clinical corollary	Failure of fusion → shaft hypospadias	The shaft–glans junction, where the mechanism changes, is the commonest site of the ectopic meatus

ORIGIN OF THE GLANULAR URETHRA — GLENISTER VS THE ENDODERMAL VIEW

	Ectodermal intrusion theory (Glenister) — outdated	Current (endodermal) view
Glanular urethra formed by	Skin (ectoderm) growing into the glans to meet the endodermal urethra	Canalisation of the endodermal urethral plate within the glans
Epithelium of the whole urethra	Mixed ectoderm and endoderm	Endoderm of the UGS throughout

- **Common ground.** On either view, **the meatus marks the ectoderm–endoderm (skin–urethral epithelium) junction**.

Prepuce

- **Tied to the urethra.** The prepuce forms with the glanular urethra and **depends on normal urethral development**.

- **Sequence:**

1. About 8 weeks: low **preputial folds** appear on each side of the shaft.
2. They join dorsally as a **flat preputial ridge** at the corona; ventrally the ridge is **blocked by the not-yet-formed glanular urethra**.
3. The folds extend **distally** over the glans and **ventrally** to fuse in the midline.
4. An epithelial **preputial lamina** separates glans mesenchyme from preputial mesenchyme.
5. **Postnatally the lamina delaminates** → preputial space → retractable prepuce.

- **The prepuce is richly innervated.**

- **Proof of the urethral link.** In hypospadias, preputial tissue is **absent ventrally and excessive dorsally** (dorsal hooded foreskin).

Penile Innervation

- **Origin.** At **week 5**, neural crest cells give rise to the peripheral and autonomic nervous systems.
- **Somatic supply: S2–S4 via the pudendal nerve** → over the sacrospinous ligament → through **Alcock's (pudendal) canal** → through the transverse perineal muscle → onto the dorsum as the **dorsal nerve of the penis**.
- **Autonomic supply** from the **vesical and prostatic plexuses: sympathetic L1–L2, parasympathetic S2–S4**.
- **Cavernous nerves** leave the pelvis **between the transverse perineal muscles and the membranous urethra**, then pass beneath the pubic arch to each corpus cavernosum.
- **Dorsal nerves:** bundled at **11 and 1 o'clock**; fan out around the tunica as far as the **junction of the corpora cavernosa with the corpus spongiosum**; leave a **nerve-free zone at 12 o'clock**.

Concept — *The penile urethra is built by two mechanisms — fusion in the shaft, canalisation in the glans — and the prepuce depends on both. So the junction between mechanisms predicts the commonest site of hypospadias, and failed ventral urethral closure predicts the hooded foreskin.*

Clinical Correlation: Hypospadias

- **Incidence:** about **1 in 250 newborn males**.
- **Definition** — **four components:** (1) **ectopic urethral meatus**; (2) **incomplete prepuce (dorsal hooded foreskin)**; (3) **ventral skin deficiency**; (4) possibly **penile curvature**.
- **Commonest meatal site:** **junction of penile shaft and glans**, matching the switch from fold fusion to direct canalisation.
- **Anatomy.** The hypospadiac penis is **normal apart from the urethra, spongiosum, overlying skin and ventral prepuce**.
- **Arrest, not deformation.** Abortive urethral and spongiosal elements (epithelium, connective tissue, smooth muscle, vessels) lie in adjacent tissue, so hypospadias is an **arrest of development, not a deformation anomaly**.
- **Aetiology:**
 - **Unknown in the overwhelming majority.**
 - Working hypothesis: **genetic susceptibility plus maternal exposure to endocrine disruptors**.
 - A known genetic defect is rare; example: **type 2 5 α -reductase deficiency** → **severe hypospadias**, because the enzyme is expressed at the fold-fusion site and is needed to fuse the groove edges.

Clinical Correlation: Correction of Penile Curvature

- **Nerve map.** Along the shaft the nerves run on the tunica albuginea and **surround the corpora ventrally up to the spongiosal junction**; from the glans to the crural separation at the pubic rami they are **bundled tightly at 11 and 1 o'clock**.
- **Plication sutures at 12 o'clock** — suitable for **dorsal plication in mild-to-moderate curvature** because it is:
 - the **nerve-free zone**;
 - where the **tunica albuginea is thickest**, so the suture anchors well.
- **Scope:** curvature with **hypospadias** and **congenital curvature without hypospadias**.

Clinical Correlation: Duplicated Urethra

- **Rare**; both **hyospadiac** and **episadiac** forms.
- **Embryology** — **longitudinal splitting of the urethral plate: complete duplication** = split from the bladder distally; **Y-duplication** = split along the shaft of the genital tubercle.
- **Signalling:** **Sonic hedgehog (Shh)** appears to control the process.

Concept — All three correlations apply one developmental map: **hypospadias is arrested development at the fusion–canalisation boundary, the safe plication site is the 12 o'clock nerve-free zone, and duplication is abnormal splitting of the single endodermal plate.**

DEVELOPMENT OF THE FEMALE EXTERNAL GENITALIA

- The clitoris forms like the penis, except that no urethra is incorporated.
- **Vestibular plate** (homologue of the urethral plate) **canalises (opening zipper)** into a wide, diamond-shaped **vestibular groove**.
- **Canalisation is androgen-independent** — it occurs in both sexes.
- **Fusion is the androgen-dependent step.** The **vestibular folds** (homologues of the urethral folds) **do not fuse**, so the groove stays open.
- **Glans clitoris** caps the narrowed paired corporal bodies; where these fuse they form a **midline septum**, starting ventrally and running about **halfway into the glans**.
- **Innervation mirrors the penis:** large bundles **densest dorsally**; **nerve-free zone at 12 o'clock**; nerves extend ventrolaterally around the tunica; the glans is supplied by **perforating branches of the dorsal nerves** at the dorsal corporoglandular junction; glanular nerve density is **lowest ventrally, beside the glans septum**.

MALE VS FEMALE EXTERNAL GENITALIA — THE GOVERNING CONTRAST

Step	Male	Female
Plate	Urethral plate	Vestibular plate
Opening zipper (canalisation)	Yes	Yes — androgen-independent
Closing zipper (fold fusion)	Yes → tubular penile urethra and raphe	No → open vestibular groove
Urethra within the organ	Penile urethra	None
Nerve-free zone	12 o'clock	12 o'clock
Orientation at 16 weeks	Penis at about 90° to the body	Clitoris recessed to the body wall

Clinical Correlation: Congenital Adrenal Hyperplasia

- **Congenital adrenal hyperplasia (CAH)** is **autosomal recessive**, **impairs cortisol synthesis**, and causes varying **virilisation of 46,XX fetuses** → genital atypia at birth.
- **Differential:** can be mistaken for **hypospadias with bilateral non-palpable testes**.
- **Mechanism:** the vestibular groove **closes to a varying degree**, forming a **clitoral-shaft urethra anatomically indistinguishable from a male urethra**.
- **Most severe form:** penis-like clitoris with **terminal meatus, glans and foreskin**.
- **Only distinguishing features:** **no palpable gonads and excess pigmentation**.

Concept — External genital sex comes down to **one androgen-dependent step, fusion of the folds**; canalisation is common to both sexes. So androgen excess in a female (CAH) closes the vestibular groove into a male-type urethra, and androgen deficiency in a male leaves it open.

KEY POINTS: EXTERNAL GENITALIA

- **Androgens start differentiation:** the ambisexual genital tubercle begins to become a penis at **8 weeks**.
- **Two-step penile urethra:** **opening zipper** (plate canalisation) → **closing zipper** (fold fusion).
- **The epithelium of the entire male urethra comes from the urogenital sinus.**
- **The clitoris is analogous to the penis** but has **no urethra**.
- **The hypospadiac penis is otherwise normal** — only urethra, spongiosum, overlying skin and ventral prepuce are abnormal.
- **12 o'clock nerve-free zone** on both penile and clitoral tunica — the site for **dorsal plication** in mild-to-moderate curvature.

Figure Legends — External Genitalia

- **Fig. 32.1** — Carnegie collection whole-mount embryos: size and complexity increase with stage.
- **Fig. 32.2** — Fetal pelvis, 9–16 weeks: after 9 weeks the **penis stands at about 90° to the body**; the **clitoris is recessed close to the body wall**.
- **Fig. 32.3** — Male vs female external genitalia, 8–16 weeks: **no vestibular fold fusion in females**.
- **Fig. 32.4** — 9-week male genital tubercle: urethral plate, and three germ layers on cross-section.
- **Fig. 32.5** — Optical projection tomography, 6.5–10.5 weeks (Li et al., 2015, *J Urol*, "double zipper hypothesis"): the **urethral plate ends within the glans**; the meatus advances **from the scrotal folds (6.5 weeks) to the proximal shaft (10.5 weeks)**; open groove clearest at 9.5–10.5 weeks with **proximal-to-distal fusion**; an epithelial tag of unknown significance.
- **Fig. 32.6** — Light-sheet microscopy at 7.5, 11 and 16 weeks: **cytokeratin 6** marks urethral plate and epidermis; **cytokeratin 7** marks urothelium; cadherin marks epithelial surfaces.
- **Fig. 32.7** — Scanning electron microscopy, 7.5–13 weeks: shaft–glans junction, distal epithelial tag, median raphe (Shen et al., 2016, *Differentiation*).
- **Fig. 32.8** — 18-week sagittal section: **remodelling of the distal urethra at the ectoderm–endoderm junction**.
- **Fig. 32.9** — Three-dimensional reconstruction of the dorsal nerve at 17.5 weeks (pubic arch, nerves, urethra, corpora, skin in layers).
- **Fig. 32.10** — Confocal imaging at 10 weeks: **CD31** (vessels) and **PGP9.5** (nerves) show the dense network at the fusion site.

- **Fig. 32.11** — (A) 24 weeks: classic distal hypospadias with dorsal hooded foreskin; (B) 23 weeks: **chordee without hypospadias**, due to penoscrotal webbing.
- **Fig. 32.12** — Schematic of **12 o'clock plication** in the nerve-sparse zone.
- **Fig. 32.13** — Optical projection tomography of the clitoris, 8–19 weeks: **opening zipper but no closing zipper**.
- **Fig. 32.14** — Light-sheet microscopy, 11-week penis and clitoris: cytokeratin 7 also marks the dorsal vestibular groove; canalisation proximal → distal in both sexes; **fold fusion in the male only**; epithelial tag in both.
- **Fig. 32.15** — Three-dimensional reconstruction of the clitoris at 24 weeks (Baskin et al., 1999, *J Urol*): **few nerves at the top and bottom of the midline**.

DEVELOPMENT OF THE HUMAN PROSTATE

Onset, Evidence Base and Stages

- **First sign.** At **10 weeks**, prostatic buds emerge from urogenital sinus epithelium (UGE) **caudal to the bladder base**.
- **Evidence base.** Human fetal data are **sparse**; mechanism and endocrinology come mainly from **animal studies** — **Jost** and **Price** established androgen dependence, later confirmed in humans.
- **Six stages, the same in human and mouse:** prebud UGS → **solid epithelial buds** from UGE → bud elongation and branching → **canalisation** of solid ducts → differentiation into **luminal and basal cells** → **secretory cytodifferentiation**.
- **Endodermal origin.** The prostate develops **below the bladder** from **FOXA1-positive UGE** (FOXA1 marks the endodermal lineage).

Verumontanum — a Uniquely Human Feature

- **What it is:** a cranio-caudally elongated hillock on the **dorsal wall of the UGS**.
- **Opening onto its apex:** the **Wolffian (mesonephric) ducts** and the **fused Müllerian ducts (prostatic utricle)**.
- **Significance:** the **interface between mesodermal epithelium** (Wolffian ducts, utricle) **and endodermal UGE**.

Budding and Branching

- **Timing:** buds at **10 weeks** (reported range **9.5–10 weeks**, because fetal age is hard to estimate).
- **Site:** most buds emerge in the **gutters lateral to the verumontanum** — as in the adult, where the ducts open lateral to the verumontanum.
- **Budding is stochastic** (mouse data, apparently true in humans): cords of UGE grow in, then regress. Evidence: **fetal ventral buds are not represented in the adult gland**.
- **Branching.** From **about 11 weeks**, growth and branching follow a **fixed spatial pattern** that **sets up the adult zonal anatomy**.
- **Canalisation** runs **proximal (urethral end) → distal (ductal tips)**, together with **luminal and basal cell differentiation**.

TABLE 32.1 MARKER EXPRESSION PROFILES IN UROGENITAL SINUS EPITHELIUM (UGE), SOLID DUCTS, AND ADULT HUMAN PROSTATE (MODIFIED FROM WANG ET AL., 2001)

Marker group	Prebud UGE	Solid ducts	Adult luminal cells	Adult basal cells
Basal-type: TP63, KRT5, KRT14, KRT19, GSTpi	+	+	–	+
Luminal-type: KRT8, KRT18	+	+	+	–

- **What it shows:** precursor epithelium (prebud UGE and solid ducts) **co-expresses basal and luminal markers**.
- **The rule:** on differentiation, **future luminal cells keep luminal markers and lose basal ones; future basal cells do the reverse**.
- **The trap:** TP63 and KRT5/14 are **basal**; KRT8/18 are **luminal**; KRT19 and GSTpi go with the **basal** set in the adult despite being present in every precursor.

Stromal Differentiation

- **Urogenital sinus mesenchyme (UGM)** differentiates into **smooth muscle**, the main component of adult prostatic stroma.
- **Direction:** ventral first, then lateral and dorsal.
- **At 19 weeks:** ductal branching is **most intense near the capsule**, where smooth muscle is most abundant.

Endocrinology

- **Androgens are essential.** Fetal testicular androgens act through the **nuclear androgen receptor (AR)**, activated by testosterone or DHT.
- **Evidence: no prostate when the AR is non-functional** (mice and humans); the **female UGS forms a prostate when exposed to androgen**.
- **DHT is critical**, although testosterone binds the AR directly: DHT is made **within the UGS by type 2 5 α -reductase (5 α R2)**, mainly in the **prostatic mesenchyme**.
- **Evidence for 5 α R2:** 5 α R2 deficiency → **rudimentary prostate** (and abnormal male external genitalia); mice lacking both 5 α R1 and 5 α R2 have smaller prostates and seminal vesicles.
- **5 α R1's role is unclear** — the 5 α R1 gene is normal in 5 α R2-deficient patients.

TESTOSTERONE VS DHT IN PROSTATIC DEVELOPMENT

	Testosterone	DHT
Source	Leydig cells of the fetal testis	Local conversion in the UGS mesenchyme by 5 α R2
AR affinity	Lower	About tenfold higher
Loss-of-function phenotype	—	5 α R2 deficiency → rudimentary prostate, severe hypospadias

Mesenchymal–Epithelial Interactions

- **Androgens act through the mesenchyme.** Androgen triggers the process via the AR, but **UGM induces and specifies prostatic identity in the epithelium**.
- **Instructive induction:** UGM can induce prostate even in **embryonic or adult bladder epithelium** (rodent and human).
- **Two-way interaction:** prostatic epithelium is needed for UGM to become smooth muscle.

Oestrogen Effects

- **Oestrogens disrupt prostatic development**; the acute effect of exogenous oestrogen (e.g. diethylstilbestrol, DES) is **squamous metaplasia**.
- **Receptor: ESR1 (oestrogen receptor α)** — metaplasia occurs in wild-type and **ESR2 (oestrogen receptor β)-knockout** mice, **not** in ESR1-null mice.

TABLE 32.2 SQUAMOUS METAPLASIA IN TISSUE RECOMBINANTS COMPOSED OF WILD-TYPE (WT) AND ESR1-NULL PROSTATIC EPITHELIUM AND MESENCHYME

Mesenchyme	Epithelium	Squamous metaplasia
Wild-type	Wild-type	+
Wild-type	ESR1-null	–
ESR1-null	Wild-type	–
ESR1-null	ESR1-null	–

- **Purpose:** shows which compartment needs the receptor — **metaplasia occurs only when ESR1 is present in both stroma and epithelium**.
- **The trap:** assuming epithelial ESR1 alone suffices; **removing ESR1 from either compartment abolishes the response**.
- **Clinical corollary:** fetal prostatic squamous metaplasia is normal in the third trimester, from **high maternal and placental oestrogen**.

Concept — In prostatic development **the hormone acts through the mesenchyme and the mesenchyme instructs the epithelium**. The same two-compartment rule explains why DHT is made in the stroma, why stroma can impose prostatic identity even on bladder epithelium, and why oestrogen-induced metaplasia needs receptor in both compartments.

Development of the Human Seminal Vesicle

- **Literature is minimal** — only two human papers.
- **Origin:** a **diverticulum of the Wolffian duct** at **10–12 weeks**.
- **Growth:** elongates and folds back → **hook-like duct**, with **extensive side ducts by months 4–5**.
- **Autonomic innervation** of the muscle coat detectable by **13 weeks**.
- **Pathology:** malformation and malignancy **extremely rare**; the main malformation is a **seminal vesicle cyst**.

PROSTATE VS SEMINAL VESICLE — ORIGIN

	Prostate	Seminal vesicle
Epithelial origin	Endodermal UGE (FOXA1-positive)	Mesodermal Wolffian duct
Onset	10 weeks (buds)	10–12 weeks (diverticulum)

KEY POINTS: PROSTATE DEVELOPMENT

- **Androgens from the fetal testis** are critical for prostatic development.
- **Mesenchymal–epithelial interactions** are fundamental, although the whole process is triggered by androgen acting through the AR.

Figure Legends — Prostate

- **Fig. 32.16** — Posterior whole mounts of bladder and prostate, 10–21 weeks: the **vas lies medial to the ureters**; the prostate is a **bulge just below the ureteric insertion**, enlarging with age.
- **Fig. 32.17** — Derivation: from **endodermal UGE** — bladder, urachus, prostate, urethra, bulbourethral glands; from the **mesodermal Wolffian duct** — ureter, vas deferens, seminal vesicle.
- **Fig. 32.18** — 9-week prebud UGS at the verumontanum: Wolffian ducts flank the Müllerian-derived utricles, both opening at the apex; FOXA1 staining.
- **Fig. 32.19** — Prostate at 12–13 weeks: utricles in the dorsal midline; ducts emerge **from the lateral and posterolateral verumontanum** (Timms, 2008; Timms and Hofkamp, 2011; Cunha et al., 2018).
- **Fig. 32.20** — Adult prostate: duct openings in the **gutters lateral to the verumontanum**; utricles and ejaculatory ducts join the urethra; **mucosal glands arise from the ventral urethra** (Timms, 1997; Cunha et al., 2018).
- **Fig. 32.21** — Ductal length from urethra to first branch point rises **from about 250 µm at 14 weeks to about 1000 µm at 17 weeks** (Cunha et al., 2018).
- **Fig. 32.22** — α -actin time course of smooth muscle: **9 weeks** sparse ventrolateral cells; **15 weeks** ventral bundles, minimal dorsal staining; **19 weeks** peripheral smooth muscle where ducts branch.

DEVELOPMENT OF THE GONADS

The Indifferent Gonad

- **Testes make sperm, ovaries oocytes, but development begins identically in both sexes.**
- **Germinal ridge** forms at **4 weeks** on the medial side of the urogenital ridge (which contains the mesonephric duct and tubules).
- **Four components: coelomic epithelium** of the germinal ridge; underlying **mesenchyme**; adjacent **mesonephros**; **primordial germ cells (PGCs)**.
- **Shape.** Gonads are **macroscopically identical at 7–8 weeks**; thereafter **testes become spherical** and **ovaries stay oblong (and longer)**.

Primordial Germ Cells and Sex Cords

- **PGC migration:** yolk sac wall, week 3 → **dorsal mesentery of the hindgut** → **gonadal ridges**.
- **Primary sex cords.** In **week 6**, surface epithelium invades the ridge mesenchyme; in humans these **do not form the seminiferous cords**, which appear to come from the mesonephric tubules.
- **First sex-specific histology: testicular cords at about 6 weeks**; ovary at **about 8 weeks**, when it segregates into a **cortical zone (ovigerous cords)** and a **medullary zone**.
- **Origin of the ovarian zones — competing views:** two successive proliferations of germinal epithelium, or ovigerous cords derived from mesonephric tubules.

Testicular Formation

- **SRY-dependent step.** Under the SRY gene product, medullary cells coalesce into **seminiferous cords of Sertoli cells enclosing the PGCs** — the testicular medulla is a zone of **mesenchymal-to-epithelial transition**.
- **Sertoli cells differentiate only if SRY protein is present**; without it the gonad becomes an ovary by default.

- **Direct Sertoli–germ cell contact** soon after PGC arrival is thought **key to normal male gamete development**.
- **Cords canalise into seminiferous tubules**.
- **Ducts**: the **rete testis** develops with the cords and at the hilum joins **5–12 residual mesonephric tubules**, which become the **efferent ductules** to the epididymal head. The **full connection** (tubules → rete → efferent ductules → epididymis) is **not established until the third month**.
- **Epididymis and vas deferens** derive from the **Wolffian (mesonephric) duct**.
- **Shape change** from oblong to round **reduces contact with the mesonephros**; a thick **tunica albuginea** forms beneath the surface mesothelium.

Müllerian-Inhibiting Substance

- **Müllerian-inhibiting substance (MIS; anti-Müllerian hormone, AMH)** is a glycoprotein from **Sertoli cells** that makes the **paramesonephric (Müllerian) ducts regress between weeks 8 and 10**.
- **Male Müllerian remnants**: **appendix testis** (small epithelial cyst at the upper pole) and **prostatic utricle** (pouch at the apex of the verumontanum).
- **In females MIS is absent**, so the Müllerian ducts persist and form **most of the female internal genitalia**.

Leydig Cells and Wolffian Derivatives

- **Leydig cells** differentiate from genital-ridge mesenchyme in response to SRY and **begin testosterone production at 8 weeks**.
- **Control of testosterone secretion**: **placental chorionic gonadotrophin early → fetal pituitary gonadotrophins later**.
- **Weeks 8–12**: testosterone turns the Wolffian ducts into paired **epididymis, vas deferens and seminal vesicle**.
- **Remnants**: cranial Wolffian duct → **appendix epididymis**; degenerating mesonephric tubules near the lower pole → **paradidymis**.

MIS VS TESTOSTERONE — THE TWO TESTICULAR HORMONES

	MIS (AMH)	Testosterone
Cell	Sertoli	Leydig
Target	Müllerian (paramesonephric) duct	Wolffian (mesonephric) duct, UGS, genital tubercle
Effect	Regression (weeks 8–10)	Stabilisation and differentiation (weeks 8–12) → epididymis, vas, seminal vesicle
Male remnants	Appendix testis, prostatic utricle	Appendix epididymis, paradidymis (Wolffian structures left undifferentiated)
If absent	Uterus and tubes persist	Wolffian ducts regress

Ovarian Formation

- **Why the ovary is the default**: **no Y chromosome → no SRY → no Sertoli-cell cords, no early MIS, no Leydig cells, no androgen → female development ensues**.
- **First ovary-specific event**: **cortex–medulla segregation at about 8 weeks**; the cortex fills with **anastomosing cortical cords**.

- **Origin of cortical cords disputed** — mesonephric tubules, or gonadal-ridge epithelial-like cells at or near the surface that become somatic ovarian cells. Either way the cords **break into clusters that the germ cells enter**.
- **Second ovary-specific event: germ cell nests at 10–11 weeks** — clusters of 2–4 germ cells joined by **intercellular bridges**; nests are **clonal** (one nest may arise from a single PGC); they **enlarge by proliferation**, then **break up by germ-cell apoptosis**.

Folliculogenesis — sequence:

1. **FOXL2-positive** somatic interstitial cells surround single germ cells → **primordial follicle** (oocyte with a single layer of **squamous** follicular cells).
 2. **About 20 weeks**: follicular cells become **cuboidal** → **primary follicle**.
 3. **At term**: granulosa proliferation → **secondary follicle** with **multiple granulosa layers**.
 4. **Postnatally**: **antral follicles** (fluid-filled cavity) appear, and the vascular **theca** organises around the follicle — it supports the follicle, **nourishes the avascular granulosa** and **synthesises sex steroids**.
- **Meiotic arrest**. Oogonia enlarge and **enter meiosis I** as **primary oocytes in fetal life**, then **arrest until puberty**; thereafter individual oocytes resume under the **monthly gonadotrophin surge**.

TESTIS VS OVARY — TIMELINE AND DEFAULT

	Testis	Ovary
Trigger	SRY → Sertoli cells	Absence of SRY (default pathway)
First specific feature	Testicular cords, about 6 weeks	Cortex and medulla segregate, about 8 weeks
Second feature	Leydig cells and testosterone, 8 weeks	Germ cell nests, 10–11 weeks
Germ cells	Enclosed by Sertoli cells in cords → seminiferous tubules	Nests → primordial (squamous) → primary (cuboidal, ~20 weeks) → secondary (term) follicles
Shape	Spherical	Oblong, longer
Hormones in fetal life	MIS and testosterone	None required for duct development

Concept — Gonadal sex is a **single switch, SRY in the pre-Sertoli cell**. Everything male follows from Sertoli cells — MIS and Leydig-cell androgen — and everything female from their absence; the male phenotype is active and hormone-driven, the female internal phenotype the default.

TABLE 32.3 EMBRYOLOGIC ORIGIN OF UROGENITAL STRUCTURES (MODIFIED FROM MOORE, PERSAUD AND TORCHIA, 2016)

Embryonic structure	Male derivative	Female derivative
Indifferent gonad (cortex / medulla)	Testis; seminiferous tubules; rete testis (mediastinum testis)	Ovary; ovarian follicles; rete ovarii (hilum, analogous to rete testis)
Gubernaculum	Gubernaculum testis (descent into the scrotum)	Ovarian ligament and round ligament of the uterus
Mesonephric tubules (mesonephros, second kidney)	Efferent ductules ; paradidymis (vestigial tubules above the epididymis)	Epoöphoron , paroöphoron (non-functional, between ovary and tube)
Wolffian (mesonephric) duct ; stalk of the ureteric bud	Appendix of epididymis; ductus deferens ; ejaculatory duct and seminal vesicle ; ureter, pelvis, calyces, collecting tubules	Gartner duct (anterosuperior vaginal wall); ureter, pelvis, calyces, collecting tubules

Embryonic structure	Male derivative	Female derivative
Intermediate mesoderm (metanephric blastema)	Kidney (metanephros); nephrons (glomerulus, proximal tubule, loop of Henle, distal tubule)	Same
Müllerian (paramesonephric) duct	Appendix testis; prostatic utricle	Paratubal cyst; uterine (fallopian) tube; uterus; cervix
Urogenital sinus	Bladder and trigone; urethra; prostate; verumontanum (seminal colliculus); urethral glands (Littre); bulbourethral glands (Cowper)	Bladder and trigone; urethra; vagina, hymen; urethral glands (Skene , lesser vestibular); greater vestibular glands (Bartholin)
Genital tubercle	Penis; glans; corpora cavernosa; corpus spongiosum	Clitoris; glans; corpora cavernosa; bulb of the vestibule
Urethral / vestibular folds	Ventral aspect of the penis	Labia minora
Labioscrotal swellings	Scrotum	Labia majora

- **Purpose:** a homologue map — **each primordium gives a paired male and female structure**, so a remnant or anomaly in one sex can be predicted from the other.
- **Pattern:** **Wolffian and mesonephric derivatives persist in the male and leave vestigial remnants in the female**, and the Müllerian duct shows the mirror image; **the ureter and collecting system come from the Wolffian system in both sexes**, which is why Wolffian anomalies carry renal anomalies (see OHVIRA, below).
- **Homologue pairs:** Littre ↔ Skene; Cowper ↔ Bartholin; corpus spongiosum ↔ bulb of the vestibule; rete testis ↔ rete ovarii; paradidymis ↔ paroöphoron.
- **The traps:** the **vagina sits under the UGS, not the Müllerian duct** — consistent with the UGE origin of vaginal epithelium (see Müllerian Structures); the **appendix testis is Müllerian**, whereas the **appendix epididymis is Wolffian**.

Gonadal Descent

Starting Position and the Two Ligaments

- **Start.** The urogenital ridge is **identical in both sexes at 7–8 weeks**; the undifferentiated gonad lies **beside the kidney in the upper lumbar region**, held **cranially** by the **cranial suspensory ligament (CSL)** and **caudally** by a ligament that becomes the **gubernaculum**.
- **Weeks 10–15:** the **testis reaches the future inguinal region** as the abdominal cavity enlarges; the **ovary descends only into the pelvis**.
- **Male:** **androgen causes the CSL to regress**, and the gubernaculum swells into a **plump ligament anchoring the testis at the groin** while the kidney moves cranially (tethering first observed in the 1700s).
- **Female:** the **CSL persists** and keeps the ovary near the kidney; the **gubernaculum stays rudimentary**.

Transabdominal Phase

- **Swelling reaction (gubernacular outgrowth):** the gubernaculum enlarges caudally.
- **Shortening of the gubernacular cord:** the cord is **taken up into the enlarging bulb**, positioning the testis over the inguinal ring so that **abdominal pressure can push it out**.
- **Cord transection misdirects the testis** → **contralateral inguinal canal or aberrant intra-abdominal position**.
- **Intra-abdominal pressure:** **controversial in this phase, important in the inguinal and scrotal phases**.

Inguinoscrotal Phase

- **From the seventh month** the gubernaculum bulges beyond the external ring toward the scrotum, while being **hollowed out by the processus vaginalis** (a peritoneal diverticulum), letting the testis leave the abdomen.
- **The gubernaculum migrates 3–5 cm** to the scrotum; the processus vaginalis lengthens with it.
- **Driving force:** intra-abdominal pressure, transmitted **directly** through the processus lumen and **indirectly** through the gubernacular cord.
- **Direction:** controlled by **androgen acting mostly indirectly via the genitofemoral nerve**, which releases **calcitonin gene-related peptide (CGRP)**.
- **After descent:** the **gubernacular bulb is resorbed**; the **processus vaginalis closes, then involutes completely**.

THE TWO PHASES OF TESTICULAR DESCENT

	Transabdominal phase	Inguinoscrotal phase
Timing	Weeks 10–15 (ends at about 15 weeks)	From the 7th month
Key events	CSL regression ; gubernacular swelling; cord shortening	Gubernacular migration 3–5 cm to the scrotum; elongation of the processus vaginalis; bulb regression
Androgen dependence	CSL regression androgen-dependent ; gubernacular swelling androgen-independent (occurs in androgen resistance)	Androgen-dependent — absent in gonadotrophin deficiency and complete androgen resistance; bulb regression also androgen-dependent (bulb stays enlarged in androgen resistance)
Other mediators	INSL3 (Leydig cell)	CGRP from the genitofemoral nerve
Intra-abdominal pressure	Controversial	Important
Failure →	Intra-abdominal testis	Inguinal or ectopic testis

INSL3

- **Insulin-like peptide 3**, a Leydig-cell product identified in **1993**, structurally related to **relaxin and insulin**; expressed in **fetal and adult Leydig cells** according to their differentiation.
- **Ins13-null mice: intra-abdominal cryptorchidism with no other reproductive defect** — the **phenotype of classic human cryptorchidism**; **early orchidopexy restores fertility potential** in these mice.
- **Androgen's overall role in descent remains unclear** — defective androgen production or metabolism gives variable cryptorchidism.

Ovarian Descent

- **A female gubernaculum exists**, from the lower ovarian pole to the labioscrotal folds, later passing through a fully formed inguinal canal.
- It **does not shorten** but **anchors the ovary in the pelvis** → **descent in the third month** onto the **broad ligament**.
- **Week 7:** the gubernaculum attaches to the Müllerian ducts; as they fuse into the uterovaginal canal it is **tethered to the uterus** and divided into the **ovarian ligament** (ovary → uterus) and the **round ligament** (uterus → inguinal canal → labia majora).

- **Female processus vaginalis** is normally obliterated; if **patent** → **indirect inguinal hernia**.

Concept — Descent is a **tug between two ligaments followed by a migration**: androgen removes the cranial restraint (CSL), *INSL3* builds the caudal anchor (gubernaculum), and androgen via CGRP steers the inguinoscrotal migration — so the site of an undescended testis tells you which step failed.

Clinical Correlation: Cryptorchidism

- **Incidence**: about 3% of newborn males; hydrocele and hernia are even more common.

ANOMALIES OF DESCENT — FAILED STEP → RESULT

Failed step	Result
CSL retained	Intra-abdominal testis
Gubernacular attachment abnormal	Testis in the inguinal canal or just outside it
Gubernacular attachment ectopic	Ectopic testis: perineum, femoral region, or near the penis
Step 1 fails (gubernacular migration to the scrotum)	Congenital cryptorchidism
Step 2 fails (closure of the processus vaginalis)	Hydrocele or inguinal hernia
Step 3 fails (complete involution of the processus vaginalis)	Acquired cryptorchidism (ascending testis) — a fibrous remnant stops the cord lengthening after birth

Clinical Correlation: Streak Gonad

- **Three genotypes**: 45,X (Turner syndrome); 46,XX gonadal dysgenesis; 46,XY gonadal dysgenesis.
- **Both testicular hormones absent**: no testosterone → Wolffian ducts regress; no MIS → Müllerian ducts form tubes, uterus, cervix and upper vagina.
- **Phenotype**: female-like internal and external genitalia.
- **Treatment**: hormone replacement for secondary sexual characteristics (the streak is non-functional); in 46,XY, risk of gonadoblastoma and later malignancy → streak gonads typically removed.

Clinical Correlation: Ovo-testicular Syndrome

- **Ovotestis**: a gonad with both testicular and ovarian tissue; extremely rare.
- **Both tissues can function hormonally** before and after birth; female fertility with normal oocytes is possible, but functioning adult spermatogenesis has not been reported.
- **Ducts follow the ipsilateral gonad**: Müllerian structures on a side without testicular tissue; Wolffian structures on a side with functioning testicular tissue.
- **Genitalia**: often atypical.

Clinical Correlation: Persistent Müllerian Duct Syndrome

- A 46,XY male with a uterus and fallopian tubes (hernia uteri inguinale).
- **Usual cause**: abnormal MIS receptor → no MIS action → female internal genitalia with a relatively normal male phenotype.
- **Typical presentation**: at exploration for bilateral abdominal testes.
- **Alternative cause**: immature testis, with MIS production out of phase with the Müllerian ducts' sensitivity window, normally 9–13 weeks.

DISORDERS OF GONADAL AND DUCTAL DEVELOPMENT — COMPARISON

Condition	Karyotype	Gonad	MIS	Testosterone	Internal ducts	External genitalia	Key point
Streak gonad	45,X / 46,XX / 46,XY	Streak, non- functional	Absent	Absent	Müllerian retained, Wolffian regressed	Female- like	Hormone replacement; remove streak if 46,XY (gonadoblastoma)
Ovotesticular	Not specified	Ovotestis	Testicular side	Testicular side	Follow ipsilateral gonad	Often atypical	Female fertility possible; no spermatogenesis reported
Persistent Müllerian duct syndrome (PMDS)	46,XY	Testes (often abdominal)	Produced but ineffective (receptor defect) or mistimed	Normal	Müllerian retained, Wolffian present	Male	Found at exploration for bilateral abdominal testes
CAH	46,XX	Ovaries	Absent	Excess (adrenal)	Female	Virilised; may be male-identical	Impalpable gonads and pigmentation are the clues

- **Organising pattern: MIS decides the Müllerian ducts; testosterone decides the Wolffian ducts and external genitalia; each condition is a different combination of the two signals.**

MIS ACTION WINDOW — TWO VALUES IN THE CHAPTER

Statement	Value	Why they differ
Müllerian regression under MIS (testicular formation)	Weeks 8–10	When regression is observed
Müllerian sensitivity to MIS (PMDS)	Weeks 9–13	Window in which the ducts can respond; relevant to mistimed MIS

KEY POINTS: GONADAL DEVELOPMENT

- **SRY** makes the indifferent gonad of the ambisexual embryo **differentiate into a testis**.
- **Leydig-cell androgens** drive male genital differentiation, **including the prostate**.
- **Sertoli-cell MIS** makes the Müllerian structures (uterus and fallopian tubes) **degenerate**.
- **Both gonads reach the pelvis by the third month; the testis enters the scrotum at about the seventh month**, guided by the gubernaculum.

Concept — Every correlation here follows from **two independent testicular signals acting on two independent duct systems**: remove both (streak gonad), block one (PMDS), give them unilaterally (ovotestis) or add androgen without MIS (CAH), and the phenotype can be predicted.

Figure Legends — Gonads

- **Fig. 32.23** — PGCs: (A) origin in the yolk sac wall at 3 weeks; (B) migration along yolk sac wall and hindgut dorsal mesentery to the genital ridges (Sadler, *Langman's Medical Embryology*, 1985).

- **Fig. 32.24** — Whole mounts of fetal gonads, 8–20 weeks: testes become spherical, ovaries oblong; epididymis visible.
- **Fig. 32.25** — Male and female gonadal and genital development (Larsen, *Human Embryology*, 1997): sexes identical through week 7. **Male:** SRY → medullary cords become seminiferous tubules, cortical cords regress; MIS → Müllerian regression, leaving appendix testis and utricule; appendix epididymis and paradidymis are mesonephric. **Female:** cortical cords invest the PGCs → follicles; Wolffian ducts degenerate; Müllerian ducts form tubes, uterus and upper vagina; **Wolffian remnants** are epoöphoron and paroöphoron in the broad ligament and **Gartner duct cysts in the anterolateral vaginal wall**.
- **Fig. 32.26** — Two stages of testicular descent (Hutson et al., 2015, *Pediatr Surg Int*): (A) testis held by CSL and gubernaculum; Wolffian duct forms epididymis and vas; (B) end of transabdominal phase (**about 15 weeks**), swollen gubernaculum holds testis near the future inguinal ring; (C) inguinoscrotal phase under **androgen and CGRP**, then processus closure and involution.
- **Fig. 32.27** — Embryological basis of anomalies of descent (Hutson et al., 2015): steps 1–3 as tabulated above.
- **Fig. 32.28** — Intraoperative photographs: (A) intra-abdominal testis at the internal ring; (B) high intra-abdominal testis; (C) normal ovary; (D) **streak gonads in XY gonadal dysgenesis**; (E) **ovotestis**; (F) **retained Müllerian structures**.

MÜLLERIAN STRUCTURES (FEMALE INTERNAL GENITALIA)

- **Males lack Müllerian structures because of testicular MIS.**
- **Females have no testosterone**, so the **Wolffian ducts resorb**; the Müllerian ducts persist and form the **uterine body, cervix and fallopian (uterine) tubes**.
- **In whole mounts the uterine body, cervix and vagina cannot be told apart**; the uterus enlarges with gestational age.

Development of the Human Female Reproductive Tract

Müllerian duct formation — sequence:

1. **Start of week 5:** coelomic epithelium **invaginates on the lateral surface of the urogenital ridge**; the infoldings become the **tubal ostia**.
2. The paired Müllerian (paramesonephric) ducts **grow caudally using the Wolffian ducts as guidewires** — **the Wolffian duct is essential** for Müllerian migration to the UGS.
3. The relationship changes along their length: **cranially** separated by mesenchyme → **further down** basement membranes in contact → **at the tip** epithelia in direct contact. Cell crossing between ducts was once suggested, but **mouse lineage tracing does not support it**.
4. **Weeks 7–8:** the Müllerian ducts lie **between the two Wolffian ducts** as they approach the UGS, along a route with two gentle curves (horizontal and vertical parts); their contact point is the **Müllerian tubercle**, a poorly defined term of limited value.
5. **Week 8:** the ducts **fuse in the midline** → **uterovaginal canal**, with a **midline epithelial septum**.
6. **Week 9:** the **septum disappears**.
7. **Female Wolffian ducts degenerate**, leaving **epoöphoron and paroöphoron** (broad ligament) and **Gartner's duct** (vaginal wall).

Uterine Tube

- **Origin:** the paired **cranial, unfused** Müllerian ducts; **fimbriae** from the irregular ostia.
- **At 8 weeks:** narrow, with a circular epithelial profile.

- **Later: mucosa becomes highly folded**, most in the **infundibulum and ampulla**; mesenchyme divides into **inner stroma** and **outer smooth muscle**.

Uterine Corpus and Cervix

- **Origin:** the **cranial uterovaginal canal**; the caudal canal takes part in vaginal development.
- **Septum anomalies:** the midline septum marks the fusion zone and is normally eliminated; **failure to regress → obstruction or duplication**.
- **Corpus epithelium** stays **simple columnar** and forms the **uterine and cervical glands**.
- **Cervix: endocervix** simple columnar and glandular; near the **external os** the Müllerian epithelium becomes **stratified squamous**, which also covers the **exocervix**.
- **Mesenchyme** forms the **endometrial stroma**, surrounded by **myometrium**.

Vagina — Origin of the Epithelium

- **The debate:** contributions from the UGS, Müllerian ducts or Wolffian ducts, alone or in combination.

ORIGIN OF VAGINAL EPITHELIUM — KOFF VS BULMER VS IMMUNOHISTOCHEMISTRY

	Koff	Bulmer	Immunohistochemistry (this chapter)
Proposal	Müllerian and UGE (sinovaginal bulbs), Müllerian predominance	UGE upgrowth forms the entire vaginal epithelial lining	Supports Bulmer
Method	Haematoxylin and eosin (H&E) only	H&E only	PAX2 (Müllerian marker) vs FOXA1 (endodermal UGE marker)

Sequence of vaginal development (PAX2/FOXA1 evidence):

1. **Weeks 8–9:** Müllerian ducts and uterovaginal canal **PAX2-positive**.
 2. **Caudal proliferation** of the uterovaginal canal → **solid vaginal plate**, abutting the **FOXA1-positive sinovaginal bulbs** (UGS-derived).
 3. **12 weeks:** plate continuous with the uterine end; mostly **PAX2-positive**; **FOXA1** limited to caudal plate, urethra and bladder.
 4. **Weeks 14–21:** plate lengthens and **FOXA1-positive UGE progressively replaces PAX2-positive Müllerian epithelium**.
 5. **Canalisation** starts at **about 15 weeks, mostly complete by 18–19 weeks**: fornices defined, **exocervix projects into the vault**; FOXA1 has reached **mid-vagina**, PAX2 runs continuously from tube to upper vagina.
 6. **21 weeks:** epithelium **multilayered and glycogenated** (likely **endogenous oestrogen**); **FOXA1 from introitus to cervix**; **PAX2 over tube, uterus and cervix**; markers switch **abruptly at the vaginal–cervical junction**.
- **Conclusions:** the entire human vaginal epithelium is **UGE-derived**; the **hymen** is FOXA1-positive and presumed **UGE-derived**.

HUMAN VS MOUSE VAGINA

	Human	Mouse
Final vaginal epithelium	UGE (FOXA1) , replacing Müllerian epithelium cranially up to the cervix	Müllerian (PAX2) , replacing the endodermal sinus vagina postnatally
Terminology	—	"Müllerian vagina" (cranial) and "sinus vagina" (caudal)
Direction of replacement	Sinus replaces Müllerian	Müllerian replaces sinus

UPPER VAGINA — TWO STATEMENTS IN THE CHAPTER

Statement	Where	What it refers to
Müllerian ducts give rise to the "upper vagina"	Fig. 32.25 legend; streak gonad text	Classic teaching: Müllerian (uterovaginal canal) origin of the upper vaginal plate
Entire vaginal epithelium is UGE-derived	Vagina text, Key Points, Table 32.3	The final epithelial lining , after UGE replaces Müllerian epithelium

- **Reconciliation:** the **upper vagina starts as Müllerian epithelium and is re-lined by UGE**; the chapter's conclusion concerns the **epithelium**.

Concept — *The female tract is built by Müllerian ducts that depend on the Wolffian duct to reach the UGS, then fuse, then lose their septum; each step has a failure mode, and the vagina is the exception whose lining is ultimately replaced by the sinus.*

Clinical Correlation: Vaginal Agenesis and Imperforate Hymen

- **Transverse vaginal septum:** failure of the vaginal canal to canalise → atresia or obstruction; about **1 in 80,000 females**; typically at the **junction of the middle and upper thirds**.
- **Imperforate hymen:** failure of the inferior end to open; the **commonest cause of vaginal obstruction**.

Clinical Correlation: Uterus Didelphys

- **Uterus didelphys:** failure of fusion of the caudal Müllerian ducts.
- **Unicornuate uterus:** one Müllerian duct fails to develop.

MÜLLERIAN ANOMALIES — FAILED STEP → ANOMALY

Failed step	Anomaly
One duct fails to develop	Unicornuate uterus
Ducts fail to fuse	Uterus didelphys
Septum fails to regress	Obstruction or duplication (septate malformations)
Vaginal plate fails to canalise	Transverse vaginal septum (middle/upper third junction)
Inferior end fails to open	Imperforate hymen (commonest obstruction)
Müllerian and Wolffian ducts both abnormal	OHVIRA

Clinical Correlation: Obstructed Hemi-Vagina and Unilateral Renal Anomalies

- **Obstructed hemi-vagina and ipsilateral renal anomaly (OHVIRA):** a rare combined **Müllerian and urinary tract anomaly** from abnormal development of **both Müllerian and Wolffian ducts**.
- **Presentation:** usually **after menarche** with **cyclic abdominal pain and vaginal discharge**; sometimes **acutely** with an **acute abdomen or tubo-ovarian abscess**.

- **The renal anomaly is not always agenesis** — **ipsilateral dysplastic kidneys or ectopic ureters** are reported.
- **Kidney often not seen on imaging** because of **dysplasia and atrophy** from abnormal (not absent) Wolffian duct formation — **not true agenesis**.

TRUE RENAL AGENESIS VS OHVIRA RENAL ANOMALY

	True renal agenesis	OHVIRA renal anomaly
Cause	Absent Wolffian duct, ureteric bud or metanephric blastema	Abnormal Wolffian duct with abnormal ipsilateral Müllerian duct
Kidney	Absent	Dysplastic or atrophic nubbin , may drain via an ectopic ureter
Associated finding	Absent ipsilateral hemi-trigone	Ectopic ureter may enter the obstructed hemi-vagina

- **Why it matters:** a dysplastic kidney with an ectopic ureter can cause **persistent vaginal leakage after resection of the vaginal septum** — **do not assume agenesis**.
- **Terminology:** the "RA" in OHVIRA now means "**ipsilateral renal anomalies**", not "renal agenesis".

KEY POINTS: MÜLLERIAN DEVELOPMENT

- **Immunohistochemistry supports Bulmer:** the entire human vaginal epithelium is **UGE-derived**, with no Müllerian or Wolffian contribution.
- **The midline septum** marks the fusion zone and is normally eliminated; **failure → uterine obstruction or duplication**.
- **Imperforate hymen** is the **commonest cause of vaginal obstruction**.

Figure Legends — Müllerian Structures

- **Fig. 32.29** — Female internal genitalia, 9–20 weeks: growth and complexity increase; **no landmarks separate corpus, cervix and vagina**.
- **Fig. 32.30** — Müllerian duct formation: (A) coelomic invagination beside the mesonephric tubule and Wolffian duct, with migrating PGCs; (B) Müllerian duct **lateral to** the mesonephros and Wolffian duct; (C) fimbria, with epoöphoron and paroöphoron.
- **Fig. 32.31** — Caudal Müllerian migration **on the Wolffian guidewire**: mesenchyme → basement membranes → epithelia in contact (Robboy et al., 2017, *Differentiation*).
- **Fig. 32.32** — Indifferent stage (**about 54 days, Carnegie stage 22**): changing duct relationship; **mesonephros regresses cranial → caudal** (Robboy et al., 2017).
- **Fig. 32.33** — Light-sheet microscopy at 9.5 weeks (E-cadherin): unfused cranial Müllerian ducts (future tubes); fused uterovaginal canal meeting the UGS; **septum and sinovaginal bulb (UGS derivative)**.
- **Fig. 32.34** — Ampulla at 21 weeks (α -actin): **highly folded epithelium**; inset, 9-week tube.
- **Fig. 32.35** — Vaginal development: (A) uterovaginal canal meets the sinovaginal bulb; (B) solid vaginal plate of both UGE and Müllerian epithelium; (C) **UGE replaces Müllerian epithelium up to the exocervix**.
- **Fig. 32.36–32.38** — PAX2/FOXA1 staining at 9, 12 and 21 weeks: at 12 weeks **PAX2** reaches nearly to the introitus and **FOXA1** only a short way into the plate; at 21 weeks the **cervix is PAX2-positive** (thin, stratified), vagina PAX2-negative, corpus strongly PAX2-positive, and **FOXA1 fills the vagina and stops at the cervix**.

- **Fig. 32.39 — Bulging imperforate hymen** in an 11-year-old with cyclic pain; ultrasound shows dilated vagina and uterus.
- **Fig. 32.40 — OHVIRA: ectopic ureter from a pelvic dysplastic renal nubbin** inserting into the obstructed hemi-vagina, with a septate uterus.

BLADDER: URETERAL DEVELOPMENT

- **Shared wall structure of bladder and ureter:** urothelium-lined lumen; **lamina propria** of extracellular matrix with disorganised smooth muscle; outer **thick smooth muscle bundles**; serosa.
- **Sex differences are small** — ureters and bladder develop alike and are of similar size, except that in males **the prostate develops below the ureteric insertion**, so the **prostatic urethra is larger** than the female homologue.
- **The ureterovesical junction inserts into the trigone identically in both sexes.**

Formation of Urogenital Sinus

- **Weeks 5–6:** the **urorectal septum** divides the **cloaca** into the **dorsal anorectal canal** and the **ventral UGS** (classic observations of Rathke and Tourneux).
- **Anorectal malformations** are thought to arise from the **cloacal membrane, not the urorectal septum**.
- **The UGS and its diverticulum, the allantois,** form bladder and urethra; the **allantois is the precursor of the urachal remnant**, and the early bladder is continuous with it.
- **The Wolffian ducts join the cloaca** just before it divides; **their entry point is the landmark** between the **cephalad vesicourethral canal** (→ **bladder and pelvic urethra**) and the **caudal UGS** (→ **phallic urethra in males; urethra and vaginal vestibule in females**).

Formation of Trigone

- **Weeks 5–6:** the **common excretory ducts** (mesonephric ducts distal to the ureteric bud origin) **dilate and join the UGS**.
- **Apoptosis** lets the **ureters disconnect from the mesonephric ducts and fuse with the bladder**.

ORIGIN OF THE TRIGONE — CLASSIC VS CURRENT VIEW

	Classic hypothesis	Current (mouse lineage tracing)
Fibromuscular wall of the trigone	Extension of ureteral tissue into the bladder	Mostly bladder smooth muscle ; only a minor ureteral contribution

- **Orifice migration:** the ureteral orifices are incorporated into the bladder and **migrate cranially and laterally** within the bladder base.

Weigert–Meyer Rule

- **Rule** (inferred mainly from duplex kidneys): the **upper-pole orifice rotates dorsally** relative to the lower-pole orifice and comes to lie **more caudal and medial**.

WEIGERT-MEYER RULE — LOWER-POLE VS UPPER-POLE URETER

	Lower-pole ureter	Upper-pole ureter
Bud origin when abnormal	Too low on the mesonephric duct	Too high on the mesonephric duct
Incorporation	Premature , with excess migration	Delayed or incomplete
Orifice position	Lateral (and cranial)	Caudal and medial — ectopic
Consequence	Vesicoureteral reflux (short intramural tunnel)	Ectopic insertion
Ectopic sites, male	—	Bladder neck, verumontanum, or Wolffian derivatives such as the vas deferens
Ectopic sites, female	—	Wolffian remnants (e.g. Gartner duct cyst) or vaginal vestibule

- **Ectopic vas deferens** results from anomalous Wolffian development: the **vas can join the ureter** instead of the verumontanum, so that both drain into the prostatic urethra via a common ... (the supplied text breaks off mid-sentence here, p. 607).

Concept — *Where the ureteric bud arises on the Wolffian duct fixes where its orifice ends up: **too low** → **lateral orifice** → **reflux**; **too high** → **caudal and medial orifice** → **ectopia into Wolffian territory**.*

Coverage note. Source pp. 584–607 only. Not supplied, and therefore not covered: the remainder of "Bladder: Ureteral Development" from p. 607 (including Figs. 32.41–32.44, cited in the supplied text: prostatic urethral size in male vs female, urorectal septum, anorectal malformations, and sites of ectopic ureter in females), kidney development, any later tables and Key Points, and the chapter's closing section.